

Seismic Data Analysis Report

Conditional Variational Autoencoder Based Probabilistic Ground Motion Prediction

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Abstract

This report summarizes a conditional variational autoencoder (CVAE) built for probabilistic ground-motion prediction. The notebook uses a screened NGA-style flatfile and predicts PGA together with 20 pseudo-spectral acceleration ordinates from 0.01 s to 6.0 s. The conditioning variables include magnitude, Joyner-Boore distance, depth, Vs30, fault-type indicators, and an intra- or inter-event flag. After cleaning and scaling, the model was trained in log space and evaluated in deterministic mode using the prior mean of the latent variable.

The dataset contains 74,159 usable records and no missing values in the selected predictors. The final model reaches an overall validation R^2 of 0.8266, which is strong for a multi-output seismic prediction problem. The main results show the expected physical trends: larger magnitudes increase response, larger distances reduce response, and softer sites produce stronger shaking.

1. Introduction

Ground-motion prediction is a core part of seismic hazard analysis because it links earthquake source properties, path effects, and site response to the shaking level expected at a station. Traditional ground-motion models are usually written as regression equations. They are transparent, but they may become restrictive when many spectral periods must be predicted together or when the data contain nonlinear interactions.

The notebook behind this report uses a CVAE to learn the response spectrum as a single multivariate output. This is useful because spectral ordinates at nearby periods are correlated, so predicting them together can preserve the overall shape of the spectrum better than fitting each period separately. The model is not meant to replace engineering judgment. Instead, it provides a compact data-driven summary of the patterns already present in the catalogue.

2. Dataset and preprocessing

The input data come from the screened NGA-style flatfile. The selected response variables are PGA and 20 pseudo-spectral acceleration ordinates. The main conditioning variables are earthquake magnitude, Joyner-Boore distance, hypocentral depth, Vs30, fault type, and an intra- or inter-event flag. Records with invalid values, sentinel values such as -999, or missing location fields were removed before modelling.

Data summary	Value
Records used	74,159
Selected predictors	Magnitude, Rjb, log_Rjb, log_Vs30, depth, fault type, intra/inter flag
Target outputs	PGA + 20 PSA periods
Missing values in selected fields	None

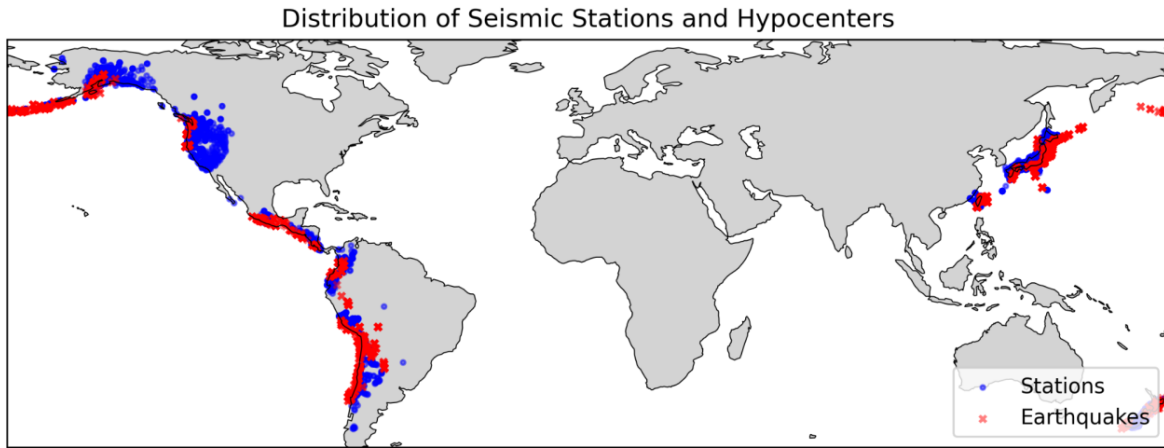


Figure 1. Geographic distribution of stations and hypocenters used in the analysis.

The map shows that the data are concentrated along active tectonic margins, especially the Pacific Rim and the western coasts of the Americas. That support is useful for learning strong-motion trends, but it also means the model is best interpreted as being strongest where the training data are dense.

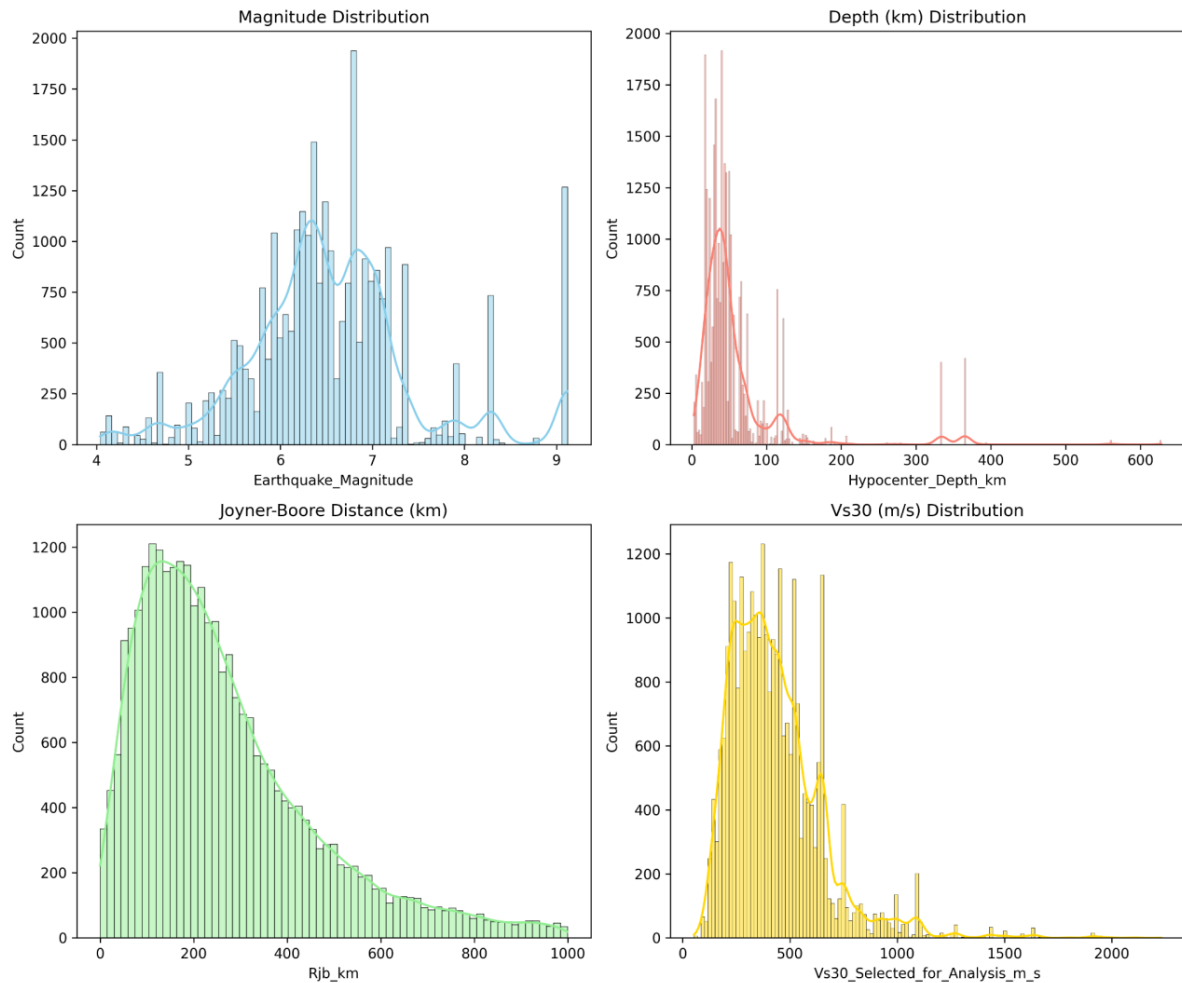


Figure 2. Distributions of magnitude, depth, Joyner-Boore distance, and Vs30.

The histograms show why scaling was needed. Magnitude is centered in the mid-6 range, depth is strongly skewed toward shallow events, distance has a long right tail, and Vs30 spans soft soil to rock-like conditions. Log transforms and standardization make these variables easier for the network to learn from.

3. Model and training setup

The CVAE is implemented in PyTorch with a compact fully connected architecture. The encoder compresses the target spectrum into a small latent Gaussian space, and the decoder reconstructs the 21 outputs from the latent sample plus the conditioning variables. The latent dimension is 8 and the hidden width is 256, which keeps the model simple enough for a tabular dataset.

Model setting	Chosen value
Hidden dimension	256
Latent dimension	8
Batch size	256
Epochs	80
Learning rate	0.0005
KL weight beta	0.5
Physics penalty weight	10.0
Dropout	0.2

The loss combines reconstruction error, KL divergence, and a soft physics penalty that discourages the response from decreasing when magnitude increases. The model is trained in standardized log space, which helps stabilize optimization and keeps the output spectrum on a smoother numerical scale. During validation, the decoder is run at the prior mean so that the reported score reflects predictive skill rather than reconstruction from the target itself.

4. Results and interpretation

The main numerical result is the validation R^2 plot. Most output periods score around 0.8 or higher, and the overall R^2 is 0.8266. That is a strong result for a multi-output seismic model because it means the network is learning a coherent spectrum rather than fitting only one easy target.

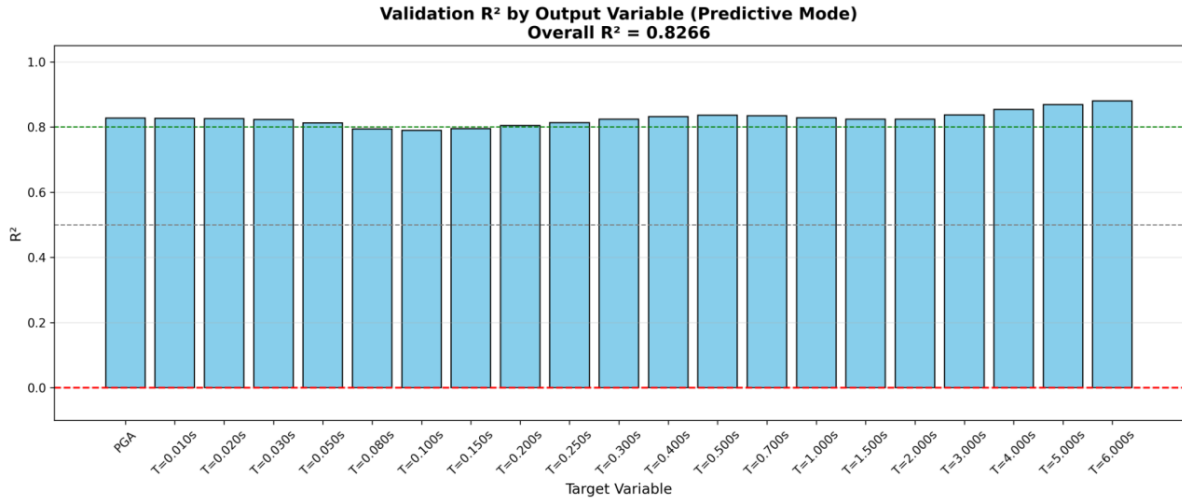


Figure 3. Validation R^2 by output variable. The overall R^2 is 0.8266.

The magnitude sweep shows the expected upward shift in spectral acceleration as M_w increases. The curves are smooth and ordered, which is important because a physically reasonable ground-motion model should increase with source size in a consistent way.

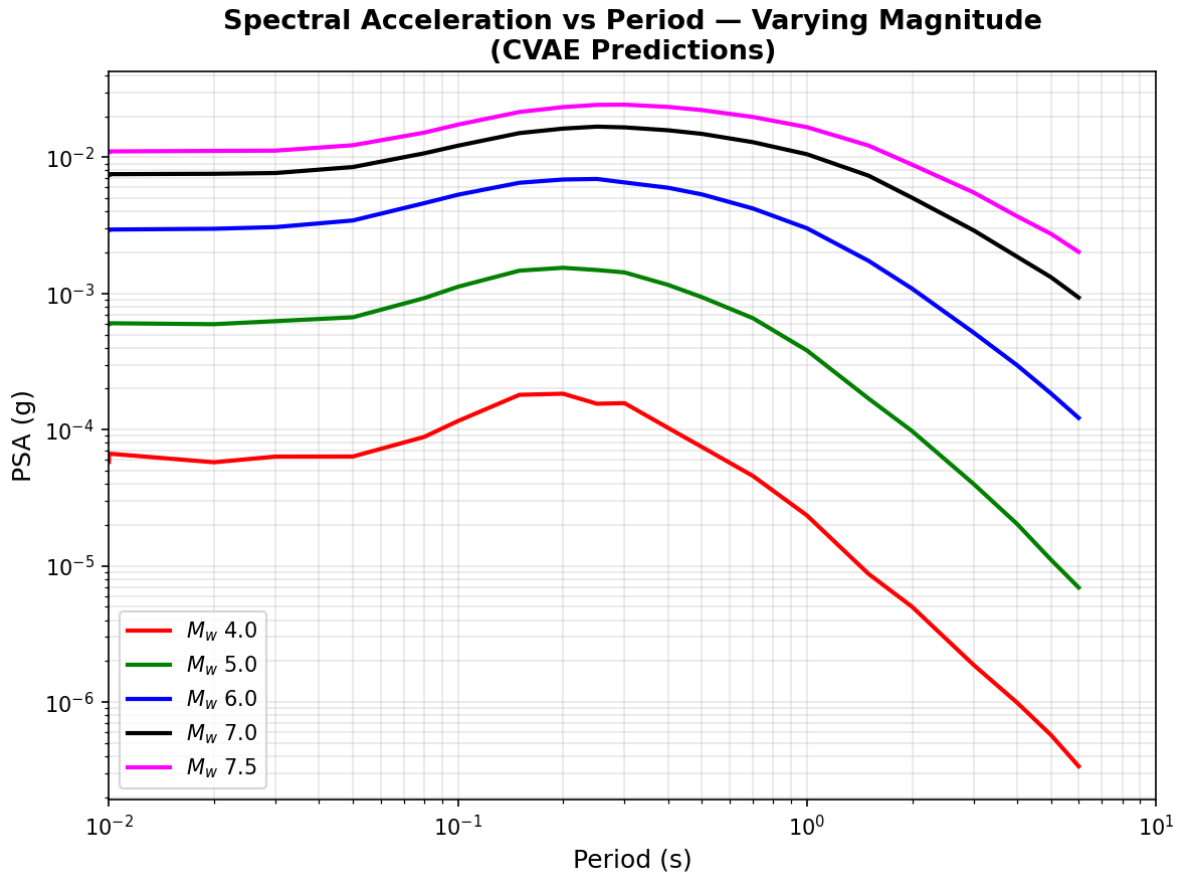


Figure 4. Spectral acceleration response for varying magnitude at fixed distance and site conditions.

The distance sweep shows the opposite trend: shaking decays as Joyner-Boore distance increases. The response is not perfectly linear, which is expected because attenuation is shaped

by geometry, path effects, and site interaction. A separate inspection of the parametric runs also shows that softer sites amplify the response more than stiff sites.

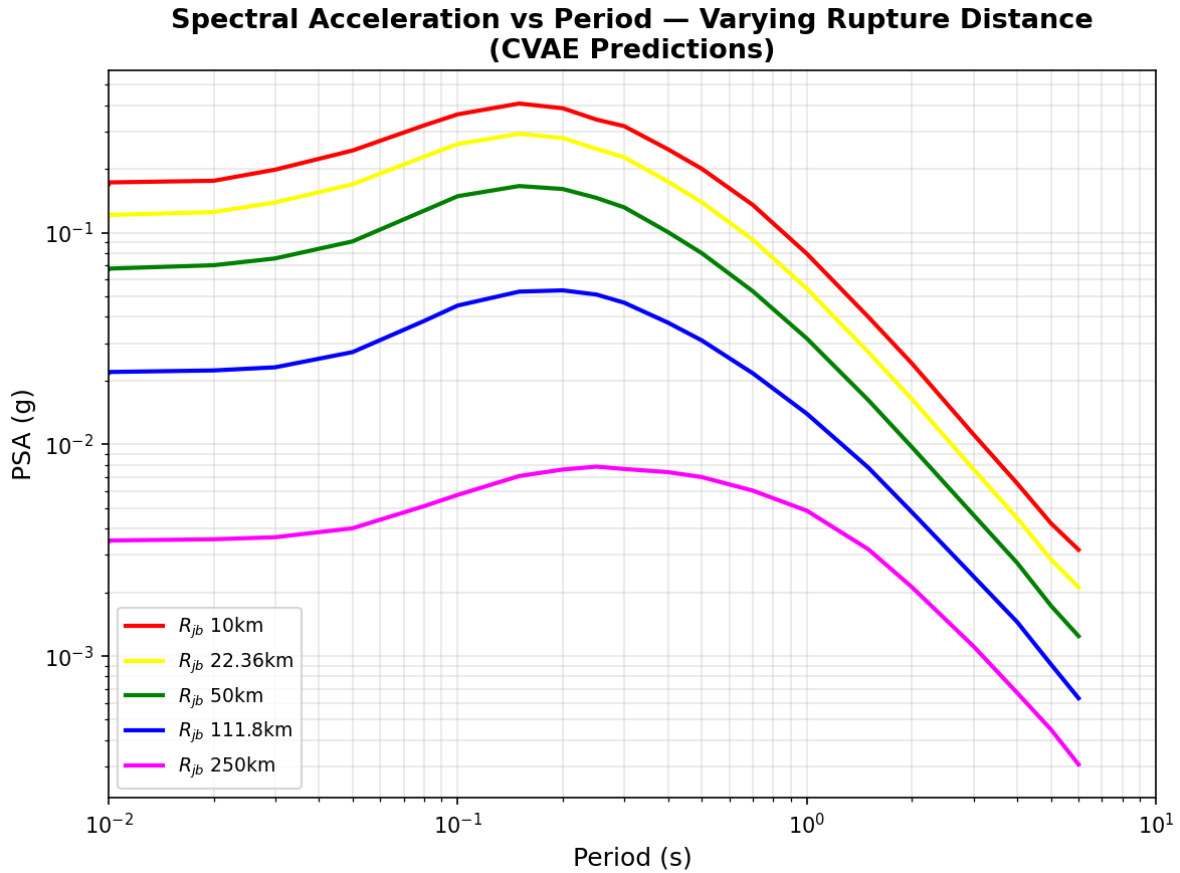


Figure 5. Spectral acceleration response for varying rupture distance at fixed magnitude and site conditions.

5. Discussion

The main value of the model is not just the R^2 score. It is the fact that the predictions follow the expected seismic trends when the inputs are changed one at a time. That makes the report easy to interpret in an engineering context. Magnitude controls the overall level, distance controls attenuation, and V_{s30} adjusts the site amplification.

The model still has limits. The catalogue is concentrated in active-margin regions, so the learned trends should not be treated as universal for every tectonic environment. Also, the latent representation captures uncertainty indirectly, so a future version could add explicit residual diagnostics or uncertainty bands.

6. Implementation summary

The notebook is easy to reproduce because the workflow is separated into a few clear steps: load the spreadsheet, clean invalid records, scale the inputs and outputs, define the CVAE, train the model, and evaluate the predictions. The core functions are small and direct, which makes the project suitable for an assignment report.

Core function	Purpose
EnhancedCVAE	Defines the encoder, decoder, and latent sampling
physics_informed_cvae_loss	Combines reconstruction, KL, and monotonicity penalty
predict_cvae_spectrum	Produces deterministic spectra for a scenario
StandardScaler + log transform	Stabilizes the learning problem

This section is useful for a report because it shows that the code is not just a black box. The model structure, loss function, and inference routine are all visible and can be explained in plain language.

7. Conclusion

This assignment shows that a CVAE can be used as a practical tool for seismic ground-motion prediction. Starting from a screened dataset of 74,159 records, the workflow cleans the data, scales the predictors, trains a compact latent model, and validates the output against the observed spectra. The final model achieves an overall R^2 of 0.8266 and reproduces the main physical trends expected in earthquake engineering.

For an assignment-level report, the method is useful because it is technically sound without being overly complicated. It demonstrates how machine learning can support seismic analysis while still preserving the main ideas of magnitude, distance, and site effects.

References

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